

Decimal Representation

- **Definition 1**

A decimal is an expression of the form

$$\pm a_0.a_1a_2a_3\cdots$$

where a_0 is a nonnegative integer and $a_1, a_2, a_3, \dots \in \{0, 1, \dots, 9\}$ are **digits**.

- **Definition 2**

Let $a_0.a_1a_2a_3\cdots$ be a decimal. If only a finite number of the digits $a_1, a_2, a_3\cdots$ are nonzero, the decimal is called **finite**. Otherwise, it is called **infinite**.

If the digits $a_1, a_2, a_3, \dots$ repeat periodically from some point on, the decimal is called a **repeating decimal**.

The periodic part of a repeating decimal is the part that repeats indefinitely, and it can be represented in the following form:

$$\begin{aligned} 0.333\cdots &= 0.\overline{3} \\ 0.454545\cdots &= 0.\overline{45} \end{aligned}$$

- **Theorem 2.1**

Let x be a nonzero decimal. If x is a finite or repeating decimal, then there exist $p, q \in \mathbb{Z}$ with $q \neq 0$ such that $qx = p$.

- **Proof 2.1.1**

Let x be a nonzero decimal with a finite or infinite periodic representation.

Consider the case where x is finite, that is,

$$x = \pm a_0.a_1a_2a_3\cdots a_n.$$

Hence,

$$10^n x = \pm (10^n a_0 + 10^{n-1} a_1 + 10^{n-2} a_2 + \cdots + 10 a_{n-1} + a_n).$$

Then, if

$$p = \pm (10^n a_0 + 10^{n-1} a_1 + 10^{n-2} a_2 + \cdots + 10 a_{n-1} + a_n) \quad \text{and} \quad q = 10^n,$$

we have that $qx = p$.

Consider the case in which x has an infinite periodic representation and assume that $x > 0$, as the case $x < 0$ is similar, that is,

$$x = a_0.b_1b_2\cdots b_m\overline{c_1c_2\cdots c_n}.$$

Hence,

$$\begin{aligned} 10^m x &= a_0b_1b_2\cdots b_m\overline{c_1c_2\cdots c_n} \\ 10^{m+n} x &= a_0b_1b_2\cdots b_m c_1c_2\cdots c_n\overline{c_1c_2\cdots c_n} \end{aligned}$$

and thus

$$\begin{aligned} 10^{m+n} x - 10^m x &= a_0b_1b_2\cdots b_m c_1c_2\cdots c_n - a_0b_1b_2\cdots b_m \\ 10^m (10^n - 1) x &= a_0b_1b_2\cdots b_m c_1c_2\cdots c_n - a_0b_1b_2\cdots b_m. \end{aligned}$$

Then, if

$$p = a_0b_1b_2\cdots b_m c_1c_2\cdots c_n - a_0b_1b_2\cdots b_m \quad \text{and} \quad q = 10^m (10^n - 1),$$

we have that $qx = p$.

◦ **Example 2.2**

(1) If $x = 0.\overline{3}$, then $10x = 3 + x$, that is, $x = \frac{3}{9} = \frac{1}{3}$.

(2) If $x = 0.\overline{45}$, then $100x = 45 + x$, that is, $x = \frac{45}{99} = \frac{5}{11}$.

◦ **Remark 2.3**

A solution of the equation $qx = p$ is represented as $\frac{p}{q}$.

• **Definition 3**

The set of **rational numbers** is

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z} \text{ with } q \neq 0 \right\}.$$

For a given $\frac{p}{q} \in \mathbb{Q}$, we say that p is the **numerator** and q is the **denominator**.

◦ **Remark 3.1**

Every integer is a rational number since $n = \frac{n}{1}$, that is, $\mathbb{Z} \subseteq \mathbb{Q}$.

◦ **Remark 3.2**

Two fractions represent the same rational number when

$$\frac{p}{q} = \frac{r}{s} \iff ps = qr.$$

◦ **Example 3.3**

$$\frac{1}{5} = \frac{2}{10} = \frac{14}{70} = \dots$$

We usually choose the representation with the smallest possible integers, that is, with the smallest possible p, q , and call this the **reduced representation**.

Since two fractions $\frac{p}{q}$ and $\frac{r}{s}$ are equal, if and only if, $ps = qr$, then the reduced representation of a rational number is that one for which $p \in \mathbb{Z}, q \in \mathbb{N}$ are such that $px + qy = 1$ for some $x, y \in \mathbb{Z}$, and we say that p, q are **coprime**.

◦ **Example 3.4 (Finite decimal number)**

The decimal 0.75 is the same as the rational number $\frac{75}{100}$.

The decimal 0.3333 is the same as the rational number $\frac{3333}{10000}$.

The decimal 0.0125 is the same as the rational number $\frac{1}{80}$.

◦ **Example 3.5 (Infinite decimal number)**

The decimal $0.333 \dots$ is the rational number $\frac{1}{3}$.

The decimal $0.454545 \dots$ is the rational number $\frac{5}{11}$.

The decimal $0.31282828 \dots$ is the rational number $\frac{3097}{9900}$.

Rational numbers are ordered as follows:

$$\frac{a}{b} < \frac{c}{d} \iff ad < bc.$$

Given $x, y \in \mathbb{Q}$, we have $x < y$, if and only if, we can find $p, r \in \mathbb{Z}$ and $q \in \mathbb{N}$ such that

$$x = \frac{p}{q}, y = \frac{r}{q} \text{ with } p < r.$$

◦ **Proposition 3.4 (Density of rational numbers)**

Between any two distinct rational numbers there is another rational number.

■ **Proof 3.4.1**

Given $x, y \in \mathbb{Q}$ with $x < y$, then

$$x < \frac{x+y}{2} < y,$$

with $\frac{x+y}{2} \in \mathbb{Q}$.

○ **Theorem 3.5**

If x is a rational number, then it can be represented as a finite or repeating decimal.

■ **Proof 3.5.1**

Assume that $x > 0$, that $x = \frac{p}{q}$ with p, q coprime and $p < q$, and let $r_0 = p$. Note that

$$10 \frac{p}{q} = \frac{10p}{q},$$

and by the division algorithm we have that

$$10p = k_1 q + r_1 \text{ with } 0 \leq r_1 < q, k_1 \in \mathbb{N}.$$

Since $p < q$, we have that

$$\begin{aligned} 10p &< 10q \\ k_1 q &\leq k_1 q + r_1 < 10q \end{aligned}$$

since $r_1 \geq 0$. Hence $k_1 < 10$. Then k_1 is a digit. Thus we obtain that

$$\frac{p}{q} = 10^{-1} k_1 + 10^{-1} \frac{r_1}{q},$$

and we compute the first digit of the decimal representation of $\frac{p}{q}$.

As $0 \leq r_1 < q$, we can repeat the same argument and obtain that

$$\frac{r_1}{q} = 10^{-1} k_2 + 10^{-1} \frac{r_2}{q}$$

with $0 \leq r_2 < q$ and $0 \leq k_2 < 10$, and we compute the second digit of $\frac{p}{q}$.

We can continue this process and obtain $r_1, r_2, r_3, \dots$ and $k_1, k_2, k_3, \dots$ such that

$$\frac{r_j}{q} = 10^{-1} k_{j+1} + 10^{-1} \frac{r_{j+1}}{q} \quad (*)$$

and

$$\frac{p}{q} = \sum_{j=1}^N 10^{-j} k_j + 10^{-N} \frac{r_N}{q}.$$

Note that there are two possibilities for the remainders $r_1, r_2, r_3, \dots$:

1. The sequence of remainders does not contain zero, so we obtain an infinite decimal representation of $\frac{p}{q}$.
2. The sequence of remainders contains zero, $r_j = 0$. From $(*)$,

$$\frac{0}{q} = 10^{-1} \cdot 0 + 10^{-1} \cdot \frac{0}{q},$$

that is, $r_{j+1} = 0$ and $k_{j+1} = 0$, therefore we obtain a finite decimal representation of $\frac{p}{q}$.

○ **Example 3.6**

$\frac{7}{12}$. We have

$$\begin{aligned}
70 &= 12 \cdot 5 + 10 \\
100 &= 12 \cdot 8 + 4 \\
40 &= 12 \cdot 3 + 4 \\
40 &= 12 \cdot 3 + 4 \\
&\vdots
\end{aligned}$$

So $\frac{7}{12} = 0.58\overline{3}$.

◦ **Theorem 3.7**

Let $\frac{p}{q}$ be a reduced representation. Its decimal representation is finite if and only if every prime factor of q is either 2 or 5.

■ **Proof 3.7.1**

($\Rightarrow$) If $\frac{p}{q}$ is finite, then $\frac{p}{q} = \frac{N}{10^k}$ for some integers N and $k \geq 0$. Hence $10^k p = qN$. Since p and q are coprime, there exist $x, y \in \mathbb{Z}$ such that

$$\begin{aligned}
px + qy &= 1 \\
10^k &= x(10^k p) + y(10^k q),
\end{aligned}$$

so $q \mid 10^k = 2^k \cdot 5^k$, which implies that the only possible prime factors of q are 2 or 5.

($\Leftarrow$) Suppose that every prime factor of q is either 2 or 5.

By the fundamental theorem of arithmetic, since $q \in \mathbb{N}$, we can express q in the following form:

$$q = 2^\alpha \cdot 5^\beta \quad \text{for some } \alpha, \beta \in \mathbb{N}_0.$$

Let $k = \max(\alpha, \beta)$. Then we can find a nonnegative integer $N' \in \mathbb{N}$ such that $10^k = q \cdot N'$.

Indeed, we have:

$$10^k = 2^k \cdot 5^k = (2^\alpha \cdot 2^{k-\alpha}) \cdot (5^\beta \cdot 5^{k-\beta}) = q \cdot (2^{k-\alpha} \cdot 5^{k-\beta}).$$

Setting $N' = 2^{k-\alpha} \cdot 5^{k-\beta} \in \mathbb{Z}$, we multiply both the numerator and the denominator of $\frac{p}{q}$ by N' , yielding:

$$\frac{p}{q} = \frac{p \cdot N'}{q \cdot N'} = \frac{p \cdot N'}{10^k}.$$

Let $N = p \cdot N'$. Since $p, N' \in \mathbb{Z}$, we have $N \in \mathbb{Z}$. Therefore, the rational number can be written as:

$$\frac{p}{q} = \frac{N}{10^k}.$$

By Definition, any rational number that can be expressed as an integer divided by a power of 10 has a decimal representation that terminates after at most k digits.

Therefore, the decimal representation of $\frac{p}{q}$ is finite.